

Grocery Delivery Analytics Pipeline

FRESHROUTE: FROM MESSY EVENTS TO DEFENSIBLE DECISIONS

FreshRoute is a fictional grocery delivery company operating 25 stores across five Ontario cities. The team built a deterministic PySpark pipeline that generates six connected tables, preserves realistic quality defects in Bronze, standardizes and validates them in Silver, and produces analysis-ready Gold tables for a native Databricks dashboard.

WHY THIS DATA MATTERS

Grocery delivery links commercial choices to fulfilment. One dashboard can compare revenue, discounts, loyalty, distance and delay.

SIX CONNECTED SOURCE TABLES

CUSTOMERS

profiles + loyalty

STORES

locations + formats

PRODUCTS

catalog + categories

ORDERS

events + payments

ORDER ITEMS

quantity + discount

DELIVERIES

distance + delay

TEAM MEMBERS

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\$172,835.74

NET REVENUE

3,636

COMPLETED ORDERS

\$47.53

AVERAGE ORDER VALUE

27.14%

ON-TIME DELIVERY

83.32%

REPEAT CUSTOMERS

From imperfect records to governed analytics

The medallion architecture separates raw evidence, cleaning rules and business logic so every metric can be traced and reproduced.



DATA MODEL AND RELATIONSHIPS

TABLE	ONE ROW REPRESENTS	KEY RELATIONSHIP
customers	Customer and loyalty profile	CustomerID -> orders
stores	Location and store format	StoreID -> orders, products
products	Store product catalogue	ProductID -> order_items
orders	Order event and payment	OrderID -> order_items, deliveries
order_items	Quantity, price and discount	OrderID + ProductID
deliveries	Driver and fulfilment outcome	OrderID -> orders

SILVER CLEANING DECISIONS

- Customers: deduplicate IDs, normalize city, loyalty and email, parse dates, reject ages outside 18-90 and median-impute age.
- Stores and products: constrain ratings, remove currency symbols, cast numbers, estimate missing cost at 65% of price and reject nonpositive values.
- Orders: normalize status and payment, parse timestamps and remove orphan customer/store references.
- Order items: standardize 0.15, 15 and 15% discount formats; reject invalid quantities and enforce product-store consistency.
- Deliveries: remove units, reject nonpositive distance or promised time, and compute delay/on-time only for delivered orders.
- Validation: assert six unique primary keys, zero orphan foreign keys and non-empty Gold outputs before dashboard use.

CRITICAL GRAIN RULE

Revenue is calculated per completed line, then summed to exactly one row per order before average order value is calculated. This prevents a line-item average from being reported as order value.

Revenue scale is healthy - context makes it useful

The Gold layer connects executive totals to category and store drivers without losing order-level accuracy.

\$172,835.74

TOTAL NET REVENUE

3,636

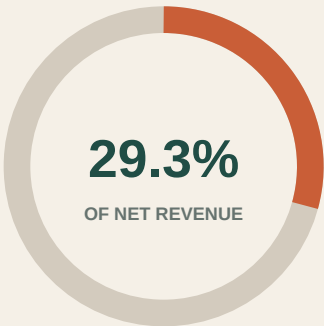
COMPLETED ORDERS

\$47.53

TRUE ORDER-LEVEL AOV

FINDING 1 | CATEGORY MIX

Meat & Seafood leads



Net revenue	\$50,649.02
Units sold	2,710
Gross profit	\$15,531.76

The category generates almost three in every ten revenue dollars. Management should compare revenue, units and gross profit together before expanding assortment because a high-ticket category can lead revenue without leading volume.

THE MANAGEMENT READING

Commercial success and operational quality should be read together. Oakville 1 leads revenue and AOV, yet only 10.59% of its analyzed deliveries are on time. The dashboard therefore pairs store revenue with service measures instead of producing a one-dimensional ranking.

HOW THE NUMBERS WERE DEFENDED

- Completed orders are distinct OrderID values after valid line items and completed status filters.
- Average order value is mean NetRevenue from orders_gold, not an average of product lines.
- Category and store metrics reconcile to the same \$172,835.74 executive total.
- Every result describes the seeded synthetic FreshRoute scenario, not a real grocery market.

Commercial conclusion

FreshRoute has meaningful demand and repeat behavior, but the best commercial segments still need operational scrutiny. Revenue alone is not the decision.

FINDING 2 | TOP STORE

Oakville 1

\$9,477.16

NET REVENUE

Completed orders

171

Unique customers

154

Average order value

\$55.42

On-time delivery

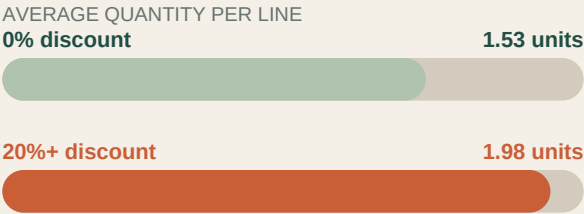
10.59%

Basket response improves; delivery reliability collapses

Two comparisons expose the main business trade-off: promotions can lift quantity, while longer travel sharply erodes service performance.

FINDING 3 | DISCOUNT RESPONSE

Higher discount, larger basket

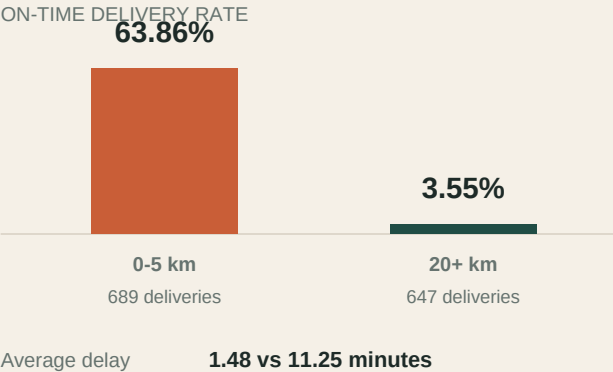


Line items	3,989 vs 879
Net revenue	\$67,574.14 vs \$14,864.28

The highest discount band raises quantity by about 29%, but it covers far fewer lines and contributes much less revenue. The result supports targeted promotions, not blanket discounting.

FINDING 4 | DISTANCE RISK

Promises fail at distance



The longest band is almost never on time. Distance-aware promise windows, smaller peak-hour delivery zones and courier-capacity monitoring are the clearest operational actions.

FINDING 5 | RETENTION VERSUS SERVICE

Customers return even while service reliability lags



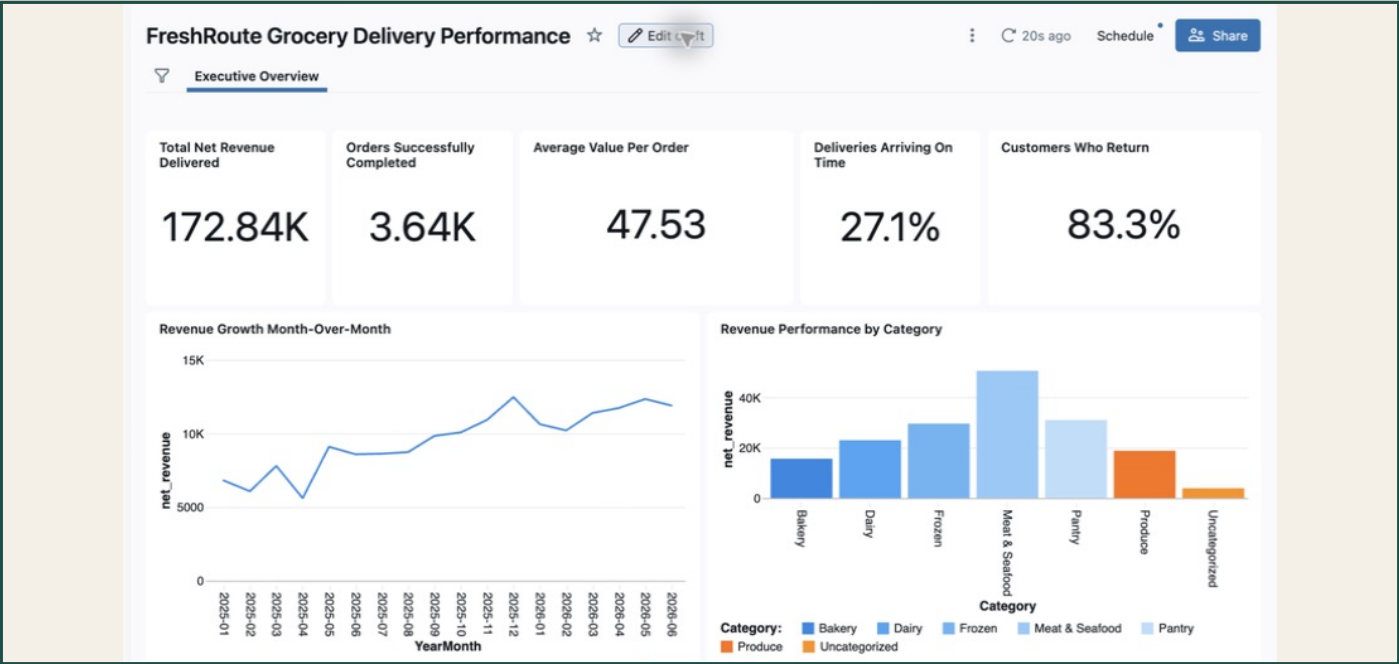
The synthetic model shows strong repurchase behavior, but this should not be treated as proof that poor delivery has no consequence. Real customer data may reveal delayed churn, complaint behavior or geographic bias that the mock data cannot capture.

INTERPRETATION DISCIPLINE

These findings validate the analytical pipeline and dashboard story. They describe engineered FreshRoute data and do not establish real-world causal effects.

A dashboard designed to move from signal to action

Executive KPIs establish scale; the next visuals explain commercial drivers, promotional response, delivery risk and customer behavior.



Published native Databricks dashboard: FreshRoute Grocery Delivery Performance | Executive Overview

DASHBOARD STORY

- Overview: five cards summarize revenue, completed orders, AOV, on-time delivery and repeat customers.
- Commercial: category, store and product views explain where revenue and profit originate.
- Action: discount, distance and loyalty visuals expose promotion, service and retention trade-offs.
- Exploration: the global Month filter changes the time context without modifying notebook code.

REFLECTION AND LIMITATIONS

The hardest engineering problem was preserving realistic foreign-key relationships while Bronze still contained testable defects. A single dependency order, deterministic seed and layer-level assertions made failures traceable. A second challenge was metric grain: revenue begins at line level, while AOV must be calculated after grouping to one order. Team integration also required preserving genuine GitHub history while harmonizing contributor notebooks into one safe, reproducible run order.

WHAT WE WOULD ADD WITH REAL DATA

- Inventory events, substitutions, weather, traffic and courier shifts.
- Automated quality expectations and scheduled dashboard refresh monitoring.
- Real distribution benchmarks, complaint outcomes and controlled promotion tests.

CONCLUSION

The project delivers a reproducible Bronze-Silver-Gold pipeline and a coherent dashboard story. Its central lesson is that validated relationships, correct aggregation grain and operational context are necessary before transaction data can support defensible business decisions.